

Clustering & Correlation: Linking Development Patterns to Economic Conditions for Renters

Ethan Shelburne

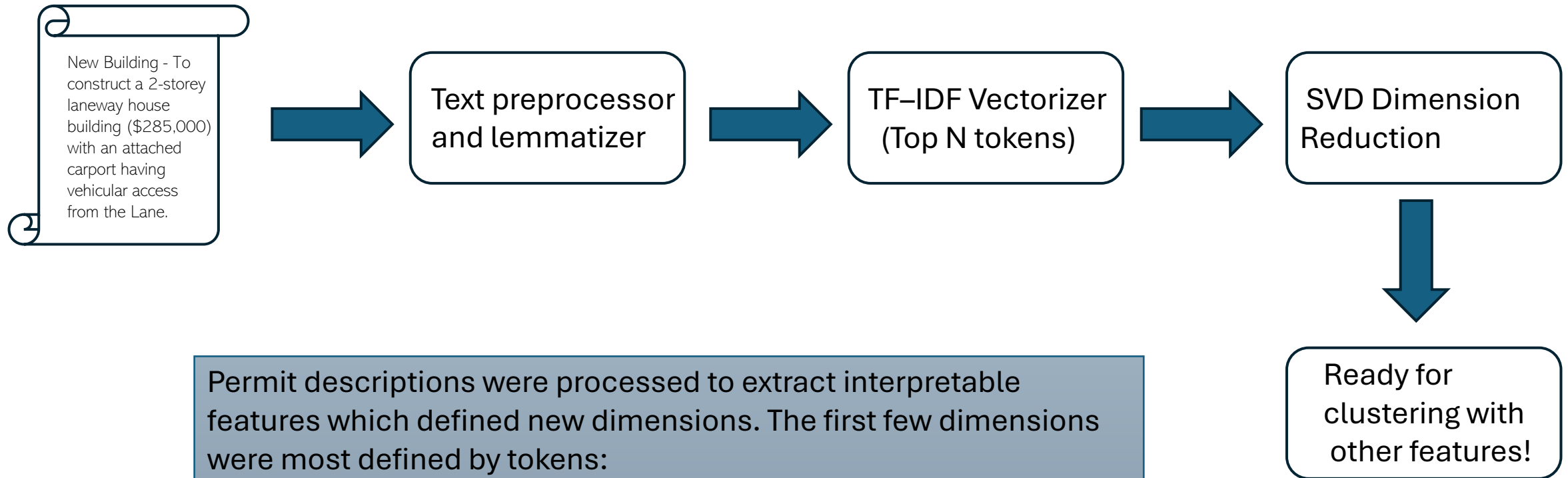
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Project Overview

- **Premise:** Vancouver faces persistent challenges around housing affordability and availability.
- **Motivation:** Development activity (builds, demolitions, renovations) may drive shifts in rent levels and vacancy rates.
- **Goal:** Use data analysis and modeling to uncover patterns in development intensity and their links to affordability metrics.
- **Data Sources:**
 - City of Vancouver Open Data → Building Permits
 - CMHC → Rental Market Survey (rents, vacancy rates)
- **Approach:** Apply NLP and clustering algorithms to uncover groups of permits, then test for correlations with economic affordability metrics.

Applying NLP to Project Descriptions



Permit descriptions were processed to extract interpretable features which defined new dimensions. The first few dimensions were most defined by tokens:

1. **“Suite / Basement / Laneway”**
2. **“Conversion / Roof / Property Line”**
3. **“Laneway House / Parking / Pad”**

Clustering Development Projects



Building permits 🏠
(25417 records,
2017-2025)

New builds 🏗️
(9464 records)

Renovations 🛠️
(7875 records)

Demolitions 🪓
(5714 records)

K-Means –
partitioning around
iteratively chosen
centroids

**Gaussian Mixture
Models (GMM)** –
probabilistic
clustering

**Agglomerative
Clustering** –
hierarchical
merging of clusters

HDBSCAN –
density based
clustering with noise
handling

Hyperparameter Optimization

Silhouette Score by Clustering Algorithm
(higher is better)

	K-Means	Gaussian Mixture	Agglomerative	HDBSCAN
New builds	0.3810	0.2754	0.4137	0.3923
Renovations	0.3837	0.3414	0.3524	0.2794
Demolitions	0.6079	0.5520	0.5895	0.7145

663 + 677 + 428 = 1768 trials	108 + 120 + 224 = 452 trials	198 + 198 + 25 = 421 trials	342 + 261 + 117 = 720 trials
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$1768 + 452 + 421 + 720 = \mathbf{3361 \text{ total trials}}$

Among several clustering metrics, we placed the most emphasis on the **silhouette score** as a balance between cohesion and separation.

Ultimately, we selected **Agglomerative (Ward) clustering** because it consistently achieved strong performance while producing reasonable numbers of **highly interpretable clusters**.



Streamlit

Interactive Web App



Live Demo: van-housing-permits-explorer.streamlit.app

Key Takeaways



- ❖ There is evidence that development patterns correlate with changes in rental affordability and accessibility.
- ❖ Clustering results reveal meaningful dynamics, e.g.:
 - ❖ Detached/duplex infill correlated with slower rent growth and lower short-term vacancies but higher large-unit vacancies in the long run.
 - ❖ Expensive family-oriented builds associated with tighter large-unit rental markets.
- ❖ Custom metric ($NF-ICF$) proved to be a statistically significant measure of development intensity.
- ❖ Vancouver has consistently become more expensive for renters since 2017, and vacancy rates have remained extremely low. As the city undergoes development initiatives like the Broadway plan, it is *essential* that research into affordability and accessibility informs all decisions.



Questions?